# Exam craft for the IMC, Kangaroo and AMC 10

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L2-ST1 Level 2 Intermediate · Strategy

How the IMC, the Grey and Pink Kangaroos and the AMC 10 are marked, when to guess, how to plan the hour, how to use the answer options (specialising, small cases, estimating, extreme cases), and how to review a paper.
**By the end you can** decide when to guess under the IMC, Kangaroo and AMC 10 marking; use the answer options to specialise, estimate, test and eliminate; plan the hour and review a past paper so that the review changes your next score.
**Prerequisites:** Thinking with algebra (L1-AL1), Factors, multiples and primes (L1-NT1). **Contests:** UKMT IMC, Grey and Pink Kangaroos, AMC 10. **Time:** about 40 minutes.

## 1 Lesson

Try this question before reading on.

*What is the value of* $\frac{3^{2026}-3^{2024}}{3^{2025}+3^{2023}}$*? (A)* 3 *(B)* $\frac{12}{5}$ *(C)* $\frac{9}{4}$ *(D)* 2 *(E)* $\frac{8}{3}$

Divide the top and the bottom by $3^{2023}$. Every exponent drops by 2023 and the value does not change, so the fraction equals

$$\frac{3^3-3^1}{3^2+3^0}=\frac{24}{10}=\frac{12}{5},$$

answer (B). The exponents are large only to slow you down, and a small version of the same question has the same answer. If your first thought was how big $3^{2026}$ is, this unit is for you.

The IMC, the Grey and Pink Kangaroos and the AMC 10 are many short questions against the clock. Ability decides most of the score, but not all of the score. A student who knows the marking, plans the hour and uses the answer options will beat a student of equal ability who does not, and these habits take a few past papers to learn.

### 1.1 The rules

**The UKMT Intermediate Mathematical Challenge.** The IMC is open to Year 11 and below in England and Wales. You have 60 minutes for 25 questions, each with five options A to E, and calculators, squared paper and measuring instruments are forbidden. Questions 1 to 15 earn 5 marks and Questions 16 to 25 earn 6 marks. A wrong answer to Questions 16 to 20 loses 1 mark, a wrong answer to Questions 21 to 25 loses 2 marks, and a blank scores 0, so the maximum is 135. The questions are roughly in order of difficulty, and UKMT says plainly that candidates are not expected to finish.

The follow-on rounds are decided by rank within each year group. Roughly the top 500 in each year are invited to the Cayley, Hamilton or Maclaurin Olympiad. Below them, about the next

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4000 in Year 9 and below are invited to the Grey Kangaroo, and about the next 15000 in Years 10 and 11 to the Pink Kangaroo. In 2026 a Gold certificate needed 61 and the Maclaurin needed 98.

**The Grey and Pink Kangaroos.** Again 60 minutes, 25 questions with options A to E, 5 marks for each of Questions 1 to 15 and 6 marks for each of Questions 16 to 25, for a maximum of 135. There is no penalty, and the paper says so on the front.

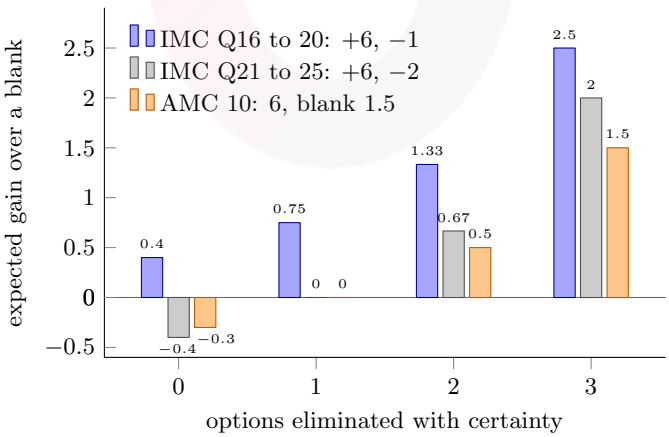
**The MAA AMC 10.** The AMC 10 is for students in grade 10 or below who are under 17.5 years old on the day. You have 75 minutes for 25 questions with five options. A correct answer earns 6 points, a blank 1.5 and a wrong answer 0, so the maximum is 150, and calculators are not allowed. The MAA invites at least the top 2.5% of AMC 10 scorers to the AIME.

Formats do change, so read the front page of your own paper before you start.

## 1.2 When to guess

**The guessing rule.**
Suppose you have eliminated $k$ of the five options with a reason, and you guess at random among the other $5-k$. The expected gain over a blank is as follows.

1. Kangaroo, and IMC Questions 1 to 15. There is no penalty, so always answer.
2. IMC Questions 16 to 20. The gain is $\frac{6}{5-k}-\frac{4-k}{5-k}=\frac{k+2}{5-k}$, which is positive even for $k=0$. Always answer.
3. IMC Questions 21 to 25. The gain is $\frac{6-2(4-k)}{5-k}=\frac{2k-2}{5-k}$. A blind guess loses 0.4 on average, one elimination breaks even, and two or more eliminations gain.
4. AMC 10. A guess is worth $\frac{6}{5-k}$ on average against a certain 1.5 for a blank. Guess once you have eliminated two options. With one eliminated the guess breaks even.




You do not need to redo this calculation in the exam. Remember the chart. On the IMC, guess freely up to Question 20, and from Question 21 guess only after two eliminations. On the AMC 10, the same two eliminations.

An elimination needs a reason, such as a parity check, a size estimate or a special case. A feeling that (E) looks wrong is not an elimination, and this is where most bad guessing comes from.



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### 1.3 Planning the hour

The IMC and the Kangaroos give 2.4 minutes per question on average and the AMC 10 gives three. Aim to spend about a minute on each early question. The time you save goes on the last ten, which are harder and, on the UKMT papers, worth more.

Work in two passes. On the first pass, answer every question you can see how to finish. If after two minutes you have no plan, mark the question and move on. On the second pass, return to the marked questions, the most promising first. Keep the last three minutes for the answer sheet and for the guessing rule.

The biggest loss on these papers is rarely a hard question you could not do. Far more often a student spends a quarter of an hour on one question, and three easier questions later in the paper are never read. Before you start a long computation, read the options, since four of them often fail a quick test.

### 1.4 Using the answer options

The options are information. Exactly one is correct, so any test that four of them fail settles the question, and you never have to prove anything. Below are the four tests I use most.

**Example 1.**
What is the largest integer that divides $n^2(n^2-1)(n^2-4)$ for every integer $n \geq 3$? (A) 60 (B) 120 (C) 180 (D) 360 (E) 720

*Solution.*
Specialise first. At $n = 3$ the expression is $9 \cdot 8 \cdot 5 = 360$, so the answer divides 360 and (E) is impossible. The other four options all divide 360, and the answer is the largest one that always works, so test 360. At $n = 4$ we get $16 \cdot 15 \cdot 12 = 2880 = 8 \cdot 360$, and at $n = 5$ we get $25 \cdot 24 \cdot 21 = 12600 = 35 \cdot 360$. The answer is (D).
For the proof, write the expression as $n \cdot (n-2)(n-1)n(n+1)(n+2)$. Five consecutive integers include a multiple of 5, a multiple of 4 and another even number, so $8 \cdot 5$ divides the product. For the factor 9, if $3 \mid n$ then $9 \mid n^2$, and otherwise two of $n-2, n-1, n+1, n+2$ are multiples of 3.
*Remark.* For a "largest divisor" question, the smallest allowed case is always the first thing to try, because the value there is an upper bound on the answer. In the exam you would stop after $n = 4$ and $n = 5$. The proof is for you, not for the answer sheet.

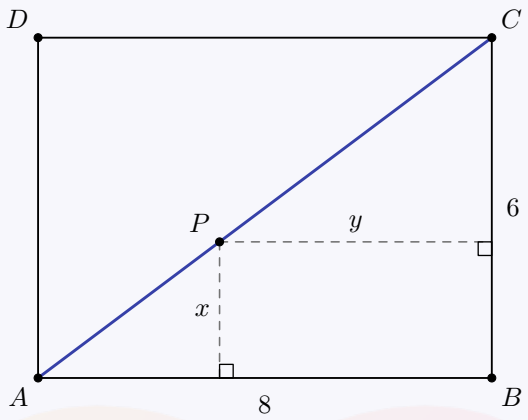
**Example 2.**
Which of the following is closest to $\sqrt{1\,002\,026} - 1000$? (A) 0.01 (B) 0.1 (C) 1 (D) 2 (E) 10

*Solution.*
The options differ by factors of about 10, so a rough estimate is enough, provided that we know how large the error can be. Squares of nearby integers give the estimate. Since $1001^2 = 1\,002\,001$ and $1002^2 = 1\,004\,004$, the value lies between 1 and 2. To decide between (C) and (D), compare with the midpoint. Since $1001.5^2 = 1\,003\,002.25 > 1\,002\,026$, the value is less than 1.5, and the answer is (C).
*Remark.* The estimate $\sqrt{a^2+b} \approx a + \frac{b}{2a}$ gives $\sqrt{1\,002\,026} \approx 1000 + \frac{2026}{2000}$, so the value is about 1.01. Use the estimate to choose, then a pair of squares to confirm. An estimate with no bound on the error is a guess.



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**Example 3.**
$ABCD$ is a rectangle with $AB = 8$ and $BC = 6$. A point $P$ lies on the diagonal $AC$, at distance $x$ from the line $AB$ and at distance $y$ from the line $BC$. What is $4x + 3y$? (A) 12 (B) 20 (C) 24 (D) 28 (E) 48




*Solution.*
The question gives no position for $P$, so $4x + 3y$ must be the same for every $P$ on $AC$. Put $P$ at an end. At $P = A$ we have $x = 0$ and $y = AB = 8$, so $4x + 3y = 24$. As a check, at $P = C$ we have $x = BC = 6$ and $y = 0$, and again $4x + 3y = 24$. The answer is (C).
For the reason, take $B$ as the origin with $A = (-8, 0)$ and $C = (0, 6)$. A point of $AC$ is $(-8 + 8t, 6t)$ with $0 \le t \le 1$, so $x = 6t$ and $y = 8 - 8t$, and $4x + 3y = 24$ for every $t$.
*Remark.* When the answer cannot depend on a free choice, make the most degenerate choice allowed, here $P$ at a vertex. Then check with a second position, since one special case can hide an error.

**Example 4.**
Five different positive integers have mean 10 and median 8. What is the largest possible value of the largest of them? (A) 26 (B) 28 (C) 30 (D) 32 (E) 34

*Solution.*
Write the numbers as $a < b < 8 < d < e$, with sum 50. To make $e$ large, make the others as small as possible, namely $a = 1$, $b = 2$ and $d = 9$. Then $e = 50 - 1 - 2 - 8 - 9 = 30$, and the five numbers $1, 2, 8, 9, 30$ satisfy every condition.
Nothing larger is possible, since $a + b + d \ge 1 + 2 + 9$ for any five such numbers. The answer is (C).
*Remark.* A "largest possible" question on a multiple choice paper is usually settled by the extreme case. Build the extreme case, check every condition (here "different" is the one people forget), and confirm in one sentence that nothing does better.

## 1.5 Checking

Repeating a calculation tends to repeat the mistake. Check by a different route instead, by putting the answer back into the question, by trying a special case, or by testing the size, the sign or the last digit. On IMC Questions 21 to 25 a wrong answer costs 8 marks relative to a correct one, so thirty seconds spent checking a doubtful answer there is well spent.



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**Common traps.**

- Stopping when one option fits a special case. Several options often agree in the first case you try, so continue until exactly one survives.
- Specialising illegally. A special case must satisfy every condition in the question, such as "different integers" or "$n \geq 3$".
- Leaving blanks on the Kangaroo or on IMC Questions 1 to 20, where a guess never loses on average.
- Guessing blind on IMC Questions 21 to 25 or on the AMC 10.
- Misreading "not", "closest to", "different" or "at least". Underline those words.
- Working an answer out on rough paper and never marking the answer sheet.

### 1.6 Reviewing a paper afterwards

Most of what you learn from a past paper comes after the hour is over. I use the following routine.

1. Mark the paper, and note your score under the real rules.
2. Sort every lost mark into one of four kinds, namely a slip, a misreading, time, or an idea you did not have.
3. Redo each question you missed, untimed, before reading any solution.
4. Read the official solutions for the last ten questions, including those you answered correctly. UKMT publishes solutions with each paper, and the AoPS wiki has several solutions to every AMC problem.
5. Keep a short log of ideas you did not have and retry those questions a fortnight later.

Slips and misreadings are fixed by checking habits. Time losses are fixed by the two-pass plan. Missing ideas are fixed by the topic units of this course, and the log tells you which ones. Be honest when sorting, since "slip" is the label everyone wants to use.

### 1.7 Which tactic to try

- The options are expressions, or the question gives no position or value for something. Specialise.
- There is a large number such as 2026. Try a small version of the question.
- The options are far apart, or the question says "closest to". Estimate and bound the error.
- The options are easy to test against the conditions. Test them, the most restrictive condition first.
- The question asks for a largest or smallest value. Build the extreme case.
- No plan after two minutes. Mark the question and move on.

## 2 Practice problems



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The problems are roughly in order of difficulty, from ⋆ to ⋆⋆⋆. Hints are in Section 4 and full solutions in Section 5.

**Problem 1** ⋆ With two minutes left in the IMC, Sam has three questions unanswered. On Question 18 Sam has eliminated nothing. On Question 23 Sam has eliminated one option with certainty, and on Question 25 two options. Sam guesses at random among the remaining options on exactly those questions where a guess increases the expected score. What is the expected gain, in marks, over leaving all three blank?

(A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) 1 (D) $\frac{16}{15}$ (E) $\frac{6}{5}$

*Tactic. Know the marking.*

**Problem 2** ⋆ Non-zero real numbers $a, b, c$ satisfy $a+b+c=0$. What is the value of

$$\frac{a^2}{bc}+\frac{b^2}{ca}+\frac{c^2}{ab}\,?$$

(A) $-3$ (B) 0 (C) 1 (D) 3 (E) 9

*Tactic. Specialise.*

**Problem 3** ⋆⋆ How many digits does $2^{100}$ have?

(A) 21 (B) 25 (C) 31 (D) 35 (E) 41

*Tactic. Estimate, and control the error.*

**Problem 4** ⋆⋆ Two circles of radii $R$ and $r$, with $R \geq r$, touch externally. A common tangent that does not pass between the circles touches them at $P$ and $Q$. Which expression equals $PQ^2$?

(A) $(R+r)^2$ (B) $4Rr$ (C) $2Rr$ (D) $R^2+r^2$ (E) $(R-r)^2$

*Tactic. Check the options in extreme cases.*

**Problem 5** ⋆⋆ A sequence has $a_1 = 2$ and

$$a_{n+1} = \frac{1+a_n}{1-a_n}$$

for $n \geq 1$. What is $a_{2026}$?

(A) $-3$ (B) $-\frac{1}{2}$ (C) $\frac{1}{3}$ (D) 2 (E) 3

*Tactic. Compute small cases.*

**Problem 6** ⋆⋆⋆ Squares $ABCD$ and $AEFG$ have sides 3 and 5 and share the vertex $A$. Both are labelled anticlockwise. What is $BG^2+DE^2$?

(A) 34 (B) 50 (C) 64 (D) 68 (E) 136

*Tactic. Specialise the position, then check with a second position.*

## 3 Contest ladder

Real past problems on the ideas of this unit, easiest first. Each source is a link to the official paper or the AoPS wiki; the note under it says where to start, not how to finish.



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1. **AMC 10A 2022 Problem 6** ⋆
   *Put in a convenient negative value of a*
2. **UKMT Grey Kangaroo 2026 Question 4** ⋆
   *Test the options, and ask first which numbers are sums of two consecutive integers*
3. **UKMT Pink Kangaroo 2026 Question 4** ⋆
   *Do the same sum up to a small number first and look at the pattern*
4. **UKMT IMC 2025 Question 16** ⋆⋆
   *Work in pence and test each option*
5. **AMC 10A Fall 2021 Problem 13** ⋆⋆
   *Ask which denominators are possible before counting anything*
6. **UKMT Pink Kangaroo 2025 Question 22** ⋆⋆
   *Write out the first five terms in terms of the second term*
7. **UKMT Pink Kangaroo 2026 Question 17** ⋆⋆
   *Try small values of b*
8. **AMC 10A 2023 Problem 16** ⋆⋆⋆
   *The number of players is a multiple of 3, so check which options can be a total*
9. **UKMT Pink Kangaroo 2025 Question 24** ⋆⋆⋆
   *The area ratio is the same for every triangle, so choose one with easy coordinates*
10. **AMC 10A 2022 Problem 22** ⋆⋆⋆
    *Count the orderings for three and four cards and look for a formula*



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## 4 Hints

1. Work out the expected gain of a random guess on each question separately, using the marks for that part of the paper.
2. The question claims that the value is the same for every allowed triple. Choose a triple with sum 0 and no zero entries.
3. Compare $2^{10}$ with $10^3$. How large can the error become after raising to the tenth power?
4. What is $PQ$ when the circles are equal? What happens to $PQ$ as the small circle shrinks towards the point where the circles touch?
5. Compute the next four terms exactly.
6. The question does not say how the squares are placed, so the value is the same for every placement. Choose the placement with $E$ on the ray $AB$.

## 5 Solutions

### Problem 1

*With two minutes left in the IMC, Sam has three questions unanswered. On Question 18 Sam has eliminated nothing. On Question 23 Sam has eliminated one option with certainty, and on Question 25 two options. Sam guesses at random among the remaining options on exactly those questions where a guess increases the expected score. What is the expected gain, in marks, over leaving all three blank?*

*(A)* $\frac{2}{5}$ *(B)* $\frac{2}{3}$ *(C)* $1$ *(D)* $\frac{16}{15}$ *(E)* $\frac{6}{5}$

Tactic. Know the marking.

Treat the three questions separately. On Question 18 a correct answer earns 6 and a wrong answer loses 1, so a blind guess gains $\frac{1}{5}\cdot 6 - \frac{4}{5}\cdot 1 = \frac{2}{5}$ on average. On Question 23 a correct answer earns 6 and a wrong answer loses 2. With four options left the guess gains $\frac{1}{4}\cdot 6 - \frac{3}{4}\cdot 2 = 0$, so the guess does not increase the expected score and Sam leaves Question 23 blank. On Question 25, with three options left, the guess gains $\frac{1}{3}\cdot 6 - \frac{2}{3}\cdot 2 = \frac{2}{3}$.

The total expected gain is $\frac{2}{5} + \frac{2}{3} = \frac{16}{15}$, answer (D).

**Answer:** (D) $\frac{16}{15}$

### Problem 2

*Non-zero real numbers $a$, $b$, $c$ satisfy $a+b+c=0$. What is the value of*

$$\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}\,?$$

*(A)* $-3$ *(B)* $0$ *(C)* $1$ *(D)* $3$ *(E)* $9$

Tactic. Specialise.

Take $(a,b,c) = (1,1,-2)$, which satisfies both conditions. The value is $\frac{1}{-2} + \frac{1}{-2} + \frac{4}{1} = 3$, answer (D). A second triple such as $(1,2,-3)$ gives $-\frac{1}{6} - \frac{4}{3} + \frac{9}{2} = 3$ again. In the exam you stop here, and the whole question takes a minute.



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For the proof, the expression equals $\frac{a^3+b^3+c^3}{abc}$. Substituting $c = -(a+b)$ gives $a^3+b^3+c^3 = a^3+b^3-(a+b)^3 = -3ab(a+b) = 3abc$, so the value is 3.

**Answer:** (D) 3

### Problem 3

*How many digits does $2^{100}$ have?*

*(A)* 21 *(B)* 25 *(C)* 31 *(D)* 35 *(E)* 41

Tactic. Estimate, and control the error.

We have $2^{10} = 1024$, so $2^{100} = 1024^{10} = 10^{30} \cdot 1.024^{10}$. The options are far apart, so we only need to know that $1.024^{10}$ lies between 1 and 10. The lower bound is clear. The upper bound is the one step that needs care, since small errors grow when raised to the tenth power. Use $1.024 < 1.1$ and $1.1^2 = 1.21 < 1.5$, so

$$1.024^{10} < 1.1^{10} = 1.21^5 < 1.5^5 = 7.59375 < 10.$$

Hence $10^{30} < 2^{100} < 10^{31}$, so $2^{100}$ has 31 digits, answer (C).

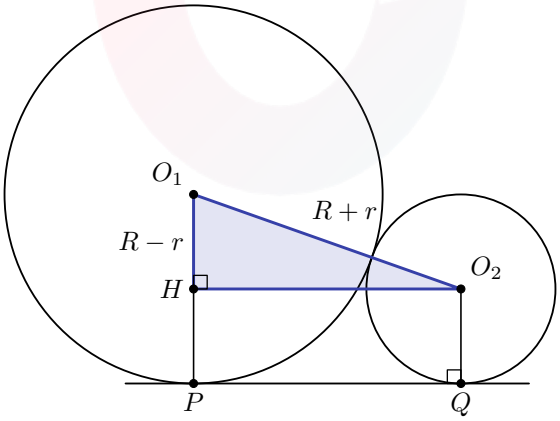
**Answer:** (C) 31

### Problem 4

*Two circles of radii $R$ and $r$, with $R \geq r$, touch externally. A common tangent that does not pass between the circles touches them at $P$ and $Q$. Which expression equals $PQ^2$?*

*(A)* $(R+r)^2$ *(B)* $4Rr$ *(C)* $2Rr$ *(D)* $R^2+r^2$ *(E)* $(R-r)^2$

Tactic. Check the options in extreme cases.




First let $R = r$. The tangent is parallel to the line of centres, so $PQ$ equals the distance between the centres, which is $2r$, and $PQ^2 = 4r^2$. With $R = r$ the options give $4r^2$, $4r^2$, $2r^2$, $2r^2$ and 0, so only (A) and (B) survive.

Next let $r$ shrink towards 0 with $R$ fixed. The small circle shrinks to the point where the circles touch, and $PQ$ tends to 0. Option (A) tends to $R^2$ and option (B) tends to 0, so the answer is (B). Two extreme cases were enough, and neither needed a diagram drawn to scale.

For the proof, let the centres be $O_1$ and $O_2$, so $O_1O_2 = R+r$. The radii $O_1P$ and $O_2Q$ are both perpendicular to $PQ$. Drop a perpendicular from $O_2$ to $O_1P$. The right-angled triangle formed has



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hypotenuse $R+r$, one leg $R-r$ and the other leg equal to $PQ$, so $PQ^2 = (R+r)^2 - (R-r)^2 = 4Rr$. This formula comes up often enough to be worth remembering.

**Answer:** (B) $4Rr$

### Problem 5

*A sequence has* $a_1 = 2$ *and*

$$a_{n+1} = \frac{1+a_n}{1-a_n}$$

*for* $n \geq 1$. *What is* $a_{2026}$*?*

*(A)* $-3$ *(B)* $-\frac{1}{2}$ *(C)* $\frac{1}{3}$ *(D)* 2 *(E)* 3

Tactic. Compute small cases.

Nothing about 2026 can be seen from the formula, so compute. The first terms are

$$a_1 = 2, \quad a_2 = \frac{3}{-1} = -3, \quad a_3 = \frac{-2}{4} = -\frac{1}{2}, \quad a_4 = \frac{1/2}{3/2} = \frac{1}{3}, \quad a_5 = \frac{4/3}{2/3} = 2.$$

Each term depends only on the previous one, and $a_5 = a_1$, so the sequence repeats with period 4. Since $2026 = 4 \cdot 506 + 2$, we get $a_{2026} = a_2 = -3$, answer (A).

(Writing $a_n = \tan\theta$, the rule says $a_{n+1} = \tan(\theta + 45^\circ)$, which explains the period 4.)

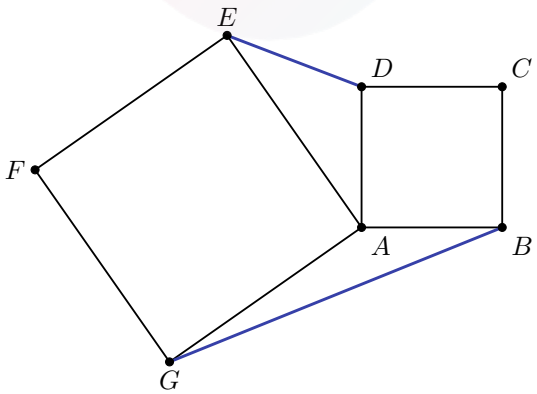
**Answer:** (A) $-3$

### Problem 6

*Squares* $ABCD$ *and* $AEFG$ *have sides* 3 *and* 5 *and share the vertex* $A$. *Both are labelled anticlockwise. What is* $BG^2 + DE^2$*?*

*(A)* 34 *(B)* 50 *(C)* 64 *(D)* 68 *(E)* 136

Tactic. Specialise the position, then check with a second position.




The value cannot depend on the placement, so put $A = (0,0)$, $B = (3,0)$, $D = (0,3)$ and the second square with $E = (5,0)$ and $G = (0,5)$. Then $BG^2 = 3^2 + 5^2 = 34$ and $DE^2 = 5^2 + 3^2 = 34$, so the sum is 68, answer (D).

As a check, rotate the second square through $90^\circ$ about $A$, so that $E = (0,5)$ and $G = (-5,0)$. Then $BG^2 = 8^2 = 64$ and $DE^2 = 2^2 = 4$, and the sum is 68 again, although each term has



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changed. The second position matters here, since in the first position the two terms are equal and an error in the setup could hide.

For the proof, use vectors from $A$. Let $\mathbf{u} = \overrightarrow{AB}$ and $\mathbf{v} = \overrightarrow{AE}$, and let $\mathbf{w}'$ denote $\mathbf{w}$ rotated anticlockwise through $90°$, so that $\overrightarrow{AD} = \mathbf{u}'$ and $\overrightarrow{AG} = \mathbf{v}'$. Rotation preserves dot products and $\mathbf{u} \cdot \mathbf{v}' = -\mathbf{u}' \cdot \mathbf{v}$. Hence

$$BG^2 + DE^2 = |\mathbf{v}' - \mathbf{u}|^2 + |\mathbf{v} - \mathbf{u}'|^2 = 2(9 + 25) - 2\,\mathbf{u} \cdot \mathbf{v}' - 2\,\mathbf{u}' \cdot \mathbf{v} = 68.$$

**Answer:** (D) 68



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